IILM UNIVERSITY

Greater Noida

QUESTION BANK

Generative AI Tools and Applications

20 Questions each for short and long questions

Short Questions (2 Marks)		CO
1.	Define Generative AI. How does it differ from traditional AI?	**1**
2.	What is a Large Language Model (LLM)? Mention any two applications of LLMs.	**1**
3	List the major stages in the evolution of Natural Language Processing (NLP).	**1**
4	What are RNNs and LSTMs? Why were they used in NLP?	**1**
5	What is a Transformer architecture? Mention its two main components.	**1**
6	Differentiate between the encoder and decoder in a Transformer architecture.	**1**
7	What is self-attention? Why is it important in Transformer models?	**1**
8	What are Query (Q), Key (K), and Value (V) in the self-attention mechanism?	**1**
9	What is multi-head attention? State one advantage of using multiple attention heads.	**1**
10	What is pre-training in LLMs? How does transfer learning help in adapting a pre-trained model to a specific task?	**1**
11	What is the main objective of a generative model?	**1**
12	How does Generative AI differ from discriminative AI in terms of output?	**1**
13	What is Natural Language Processing (NLP)? Mention one of its applications.	**1**
14	What limitation of traditional rule-based NLP led to the use of machine learning approaches?	**1**
15	What is the role of deep learning in NLP?	**1**
16	What is sequence processing in the context of NLP?	**1**
17	What is positional encoding in a Transformer?	**1**
18	Why is positional information required in Transformer models?	**1**
19	What is an attention score in a Transformer?	**1**
20	What is fine-tuning in the context of a pre-trained language model?	**1**
21	What is GPT? Mention its full form.	**2**
22	What is meant by an autoregressive language model?	**2**

23	What is the role of the decoder in GPT architecture?	**2**
24	What is next-token prediction in GPT models?	**2**
25	Mention any two variants of GPT models.	**2**
26	What is text summarization using GPT?	**2**
27	Mention two applications of GPT in natural language processing.	**2**
28	What is a dialogue system?	**2**
29	What is the role of prompts in a GPT-based chatbot?	**2**
30	Mention two features required for an e-commerce customer-support chatbot.	**2**
31	What does GPT stand for? State its basic purpose.	**2**
32	What is meant by an autoregressive language model?	**2**
33	What is next-token prediction in GPT models?	**2**
34	Why is GPT called a decoder-only Transformer model?	**2**
35	Mention any two variants or generations of GPT models.	**2**
36	What is text summarization using GPT?	**2**
37	How can GPT models be used for machine translation?	**2**
38	What is a dialogue system? Give one example.	**2**
39	What is the role of context in a GPT-based chatbot?	**2**
40	Mention any two features required in an e-commerce customer-support chatbot.	**2**
41	What are chat-based AI systems? Give one example.	**3**
42	What is a text-to-image generation model?	**3**
43	Mention any two applications of image generation tools.	**3**
44	What is AI-based audio or speech generation?	**3**
45	What is video generation using AI? Give one example.	**3**
46	What are code generation tools? Mention one example.	**3**
47	What is prompt engineering?	**3**
48	What are the main elements of an effective prompt?	**3**
49	What is prompt optimization?	**3**
50	Why is it useful to specify the desired output format in a prompt?	**3**
51	What is conversational AI?	**3**
52	What is meant by multimodal AI generation?	**3**
53	What is an image prompt?	**3**
54	Mention two factors that can affect the quality of an AI-generated image.	**3**
55	What is text-to-speech generation?	**3**
56	What is AI-based video synthesis?	**3**
57	What is code completion?	**3**
58	Why should a prompt clearly specify the task?	**3**
59	What is the role of context in prompt engineering?	**3**

60	What is iterative prompt refinement?	
61	What is the role of Generative AI in industry?	**4**
62	What is AI-driven automation? Give one example.	**4**
63	What is meant by intelligent decision-making using AI?	**4**
64	What is bias in an AI system?	**4**
65	What is fairness in AI?	**4**
66	Why is accountability important in AI systems?	**4**
67	Mention any two data privacy risks associated with Generative AI.	**4**
68	What is meant by responsible AI?	**4**
69	What is human-AI collaboration?	**4**
70	Mention any two sectors where Generative AI is being used.	**4**
71	What is intelligent automation?	**4**
72	Mention two areas where AI can automate routine tasks.	**4**
73	What is meant by transparency in AI systems?	**4**
74	Why should AI decisions be explainable?	**4**
75	What is algorithmic bias?	**4**
76	Mention two ways to protect sensitive data when using AI systems.	**4**
77	What is AI governance?	**4**
78	What is meant by human oversight in AI systems?	**4**
79	Mention two skills that may become important in an AI-enabled workplace.	**4**
80	What is an AI use-case or case study?	**4**
81	What is the role of Generative AI in industry?	**5**
82	What is AI-driven automation? Give one example.	**5**
83	What is meant by intelligent decision-making using AI?	**5**
84	What is bias in an AI system?	**5**
85	What is fairness in AI?	**5**
86	Why is accountability important in AI systems?	**5**
87	Mention any two data privacy risks associated with Generative AI.	**5**
88	What is meant by responsible AI?	**5**
89	What is human-AI collaboration?	**5**
90	Mention any two sectors where Generative AI is being used.	**5**
91	What is AI-generated content? Give one example.	**5**
92	How can Generative AI help in creating product descriptions?	**5**
93	What is an AI tutor?	**5**
94	How can Generative AI help teachers prepare learning materials?	**5**
95	What is medical image analysis?	**5**

96	How can AI help identify potential drug candidates?	**5**
97	What is the role of Generative AI in financial report generation?	**5**
98	How can AI assist in generating music?	**5**
99	What is AI-assisted code completion?	**5**
100	Mention any two software development tasks that can be supported by Generative AI.	**5**

Long Questions (6 Mark)		**CO**
1.	Explain the main stages in the evolution of NLP, from rule-based methods to Transformers.	**1**
2.	Compare traditional AI and Generative AI with suitable examples.	**1**
3	Explain the role of RNN, LSTM, and GRU in NLP. Why were they replaced by Transformers?	**1**
4	Explain the main components of a Transformer architecture and their functions.	**1**
5	Differentiate between the encoder and decoder of a Transformer with a simple example.	**1**
6	Explain self-attention using Query (Q), Key (K), and Value (V).	**1**
7	Explain how multi-head attention works. Why is it useful in language processing?	**1**
8	Explain the steps involved in pre-training an LLM on a large text dataset.	**1**
9	Compare Masked Language Modeling (MLM) and Causal Language Modeling (CLM) with examples.	**1**
10	A pre-trained LLM needs to be adapted for a specific task. Explain any three transfer-learning strategies that can be used.	**1**
11	Explain the major characteristics of Large Language Models and how they differ from conventional NLP models.	**1**
12	Explain how deep learning changed the way language-related tasks were handled compared with earlier NLP approaches.	**1**
13	Describe the limitations of sequential models that motivated the development of Transformer architectures.	**1**
14	Explain the role of positional encoding in a Transformer and why it is needed when processing a sequence.	**1**

15	Describe the complete flow of information through a Transformer encoder from input tokens to contextual representations.	**1**
16	Explain how attention allows a model to identify relationships between different words in a sentence.	**1**
17	Compare single-head attention and multi-head attention in terms of the information they can capture from a sentence.	**1**
18	Explain the difference between a model trained from scratch and a pre-trained model followed by task-specific adaptation.	**1**
19	A pre-trained language model needs to be adapted for sentiment analysis. Explain how transfer learning can be used for this task.	**1**
20	Explain the advantages and challenges of using pre-trained LLMs for multiple downstream NLP tasks.	**1**
21	**Explain the basic architecture of GPT and the role of its main components.**	**2**
22	**Explain how an autoregressive language model generates text step by step.**	**2**
23	**Compare GPT with an encoder-based model such as BERT.**	**2**
24	**Explain how GPT can be used for text generation, summarization, and translation.**	**2**
25	**Explain the working of a GPT-based dialogue system with a simple example.**	**2**
26	**Identify the main components needed to design a GPT-based chatbot. Explain their roles.**	**2**
27	**Design a simple GPT-based chatbot for e-commerce customer support. Explain its workflow.**	**2**
28	**A customer asks, "Where is my order?" Explain how a GPT-based chatbot can handle this query.**	**2**
29	**Explain how prompts can be designed to improve the responses of an e-commerce chatbot. Give suitable examples.**	**2**
30	**Discuss the benefits and limitations of using GPT for customer-support chatbots.**	**2**
31	Explain the basic architecture of GPT and describe the role of its major components.	**2**
32	Explain how an autoregressive language model generates a sentence using next-token prediction.	**2**
33	Compare GPT with an encoder-based model such as BERT in terms of architecture and text generation capability.	**2**
34	Explain how GPT can be used for text generation, summarization, and translation with suitable examples.	**2**
35	Describe the working of a GPT-based dialogue system from receiving a user query to generating a response.	**2**

36	Explain the major components required to design a GPT-based chatbot and the function of each component.	**2**
37	Design a simple GPT-based chatbot for e-commerce customer support. Explain its workflow and major features.	**2**
38	An online customer asks, “Where is my order?” Explain how a GPT-based customer-support chatbot can process and respond to this query.	**2**
39	Explain how prompt design and conversation context can improve the responses of an e-commerce chatbot. Give suitable examples.	**2**
40	Discuss the benefits and limitations of using GPT-based chatbots for e-commerce customer support.	**2**
41	**Explain the role of Generative AI in industry with suitable real-world examples.**	**3**
42	**Explain how AI-driven automation can improve business processes. Give suitable examples.**	**3**
43	**Explain how AI can support intelligent decision-making in an organization.**	**3**
44	**Explain bias, fairness, and accountability in AI systems with suitable examples.**	**3**
45	**Discuss the major data privacy and security risks when using Generative AI.**	**3**
46	**Explain any four responsible AI practices that organizations should follow.**	**3**
47	**A company uses an AI system to screen job applications. Explain how bias and fairness issues can arise and suggest suitable measures.**	**3**
48	**Explain how humans and AI can work together in the future workplace.**	**3**
49	**Select any two sectors and explain how Generative AI can be used in each sector with suitable examples.**	**3**
50	**Discuss the possible impact of Generative AI on future jobs, skills, and the workforce.**	**3**
51	**Explain the main features of chat-based AI tools and their common uses.**	**3**
52	**A user wants to create a realistic image using an AI tool. Explain the information that should be included in the prompt.**	**3**
53	**Explain how AI-based speech generation can be used to create educational or accessibility content.**	**3**
54	**Compare AI-generated video and animation based on their inputs, outputs, and common applications.**	**3**
55	**Explain how code-generation AI can help a developer write, explain, and debug a program.**	**3**
56	**Convert a vague prompt into a clear and detailed prompt for generating an educational explanation. Explain the improvements made.**	**3**

57	Analyze the effect of role, context, constraints, and output format on the quality of an AI response.	3
58	A generated response is incomplete and contains irrelevant information. Explain the prompt optimization steps that can be used to improve it.	3
59	Design a prompt that asks an AI tool to generate a Python program for a student attendance system. Specify the requirements and expected output.	3
60	Design a multimodal AI workflow for creating a short educational lesson using text, images, voice, and video generation tools.	3
61	Explain how chat-based AI tools generate responses to user prompts. Give suitable examples.	4
62	Explain how text-to-image models generate an image from a text prompt.	4
63	Compare text, image, audio, and video generation tools based on their inputs and outputs.	4
64	Explain the uses of AI tools for audio, speech, video, and animation generation.	4
65	Explain how code-generation tools can help developers in writing and debugging programs.	4
66	Write an effective prompt to generate a detailed image for a given scenario. Explain the parts of your prompt.	4
67	Compare a weak prompt and an effective prompt for generating text. Explain why the second prompt gives better results.	4
68	Explain any four techniques for optimizing prompts and improving AI-generated responses.	4
69	Design a prompt for an AI tool to create a short educational video on “Artificial Intelligence.” Mention the expected output.	4
70	A student wants to create a presentation using AI-generated text, images, voice, and video. Suggest suitable types of AI tools and explain how they can be used together.	4
71	Explain how Generative AI can automate routine tasks in an organization. Give suitable examples.	4
72	A company uses AI to approve customer loan applications. Explain how AI can support this decision-making process and what factors should be considered.	4
73	Explain how biased training data can affect the output of an AI system. Give a suitable example.	4
74	An AI system produces different results for different groups of users. Explain how fairness can be evaluated and improved.	4
75	Explain the importance of data protection when organizations use Generative AI tools.	4

76	Design a simple set of responsible AI guidelines for an organization using Generative AI.	4
77	A company introduces AI to assist employees with routine tasks. Explain how humans and AI can share responsibilities effectively.	4
78	Analyze how Generative AI may change the skills required by employees in the future workplace.	4
79	Study an AI application in healthcare, education, finance, or manufacturing and explain its problem, AI solution, and benefits.	4
80	An organization plans to introduce an AI system for customer service. Develop a simple implementation plan covering automation, human oversight, data security, and accountability.	4
81	Explain the role of Generative AI in industry with suitable real-world examples.	5
82	Explain how AI-driven automation can improve business processes. Give suitable examples.	5
83	Explain how AI can support intelligent decision-making in an organization.	5
84	Explain bias, fairness, and accountability in AI systems with suitable examples.	5
85	Discuss the major data privacy and security risks when using Generative AI.	5
86	Explain any four responsible AI practices that organizations should follow.	5
87	A company uses an AI system to screen job applications. Explain how bias and fairness issues can arise and suggest suitable measures.	5
88	Explain how humans and AI can work together in the future workplace.	5
89	Select any two sectors and explain how Generative AI can be used in each sector with suitable examples.	5
90	Discuss the possible impact of Generative AI on future jobs, skills, and the workforce.	5
91	Explain how Generative AI can create different types of marketing content for a business.	5
92	A teacher wants to create different learning materials for students with different learning levels. Explain how Generative AI can help.	5
93	Explain how Generative AI can assist doctors in analyzing medical images. Mention its benefits and limitations.	5
94	Describe the major steps involved in using AI for drug discovery.	5
95	A bank wants to use Generative AI to summarize financial reports. Explain how the system can be used and what precautions are needed.	5

96	Explain how Generative AI can support business analysts in finding insights from business data and reports.	5
97	Compare AI-generated art and AI-generated music based on their inputs, outputs, and applications.	5
98	Explain how Generative AI can assist a software developer throughout the software development life cycle.	5
99	Design a simple AI-based solution for generating personalized advertisements for different customer groups.	5
100	A company wants to automate code documentation and test-case generation using AI. Explain the proposed workflow and possible challenges.	5

Declaration:

I declare that the questions that I have contributed to the question bank are solely prepared by me. Further, I have maintained the confidentiality of the same and same will not be shared with anybody.

Signature

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QUESTION BANK VETTING REPORT

Q. No.	Level*	COs	KL (Bloom's Taxonomy)	Generated by AI (Yes/No)	Verified with Remarks

***Level: Very Easy, Easy, Medium, Difficult, Very Difficult**

Comments (Overall):

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Name & Signature of the Vetter

Approved by:

D.L. **Dean** **Director**